AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 31 · WORKBOOK

Design the Decision Process

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

One connected decision-flow record.

Name final outputs, trace candidate state, repair an OR error and design the independent room process.

Keep one habit: Keep validation, conditions, state changes and stopping rules inspectable.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your **explanation**.

- Without opening the old lesson, explain one key idea from Choose **Input**s and **Output**s. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Humans Set Boundaries. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Generate, Retrieve, or Calculate?. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ **Earlier wording, alternatives and picture transcripts**

Choose Inputs and Outputs

Humans Set Boundaries

Generate, Retrieve, or Calculate?

W1 · Name each final output and its condition

> Distinguish a rejected candidate from a completed no-match search.

Your evidence record

- Activity reference for W1–W3 and W5: valid minutes are whole 0–30; Quiet required is Yes/No. Catalogue order: B1 Bookmark drawing (10 min, quiet Yes), B2 Story circle (8 min, quiet No), B3 Pattern puzzle (12 min, quiet Yes), B4 Tower challenge (15 min, quiet No). Require duration≤available minutes AND (quiet not required OR activity quiet). Start at B1, advance after failure, stop at the first fit, or no match after all fail. Invalid/missing input needs clarification before search. A new corrected request starts a fresh run.

- Name the three kinds of final **output** in the activity **process** and the condition that leads to each. Explain why a **candidate** failing its time check is not automatically a final no-match result.

> **My reasoning and evidence**

MAIN LESSON · SECTION 13

W2 · Track the current candidate

> Keep request values fixed while candidate state advances.

Your evidence record

- Trace 9 minutes with Quiet required No. Record the current **candidate**, time result, quiet condition and next step until the **process** stops. Explain why the request values should not change as the **candidate** advances.

My reasoning and evidence

MAIN LESSON · SECTION 14

W3 · Repair OR to preserve both requirements

> Use the supplied five-minute case to expose the wrong condition.

Your evidence record

- A design says “If the activity is quiet OR its time fits, suggest it.” For 5 minutes with Quiet required Yes it selects B1. Identify the failed requirement and replace the condition with a correct one.

  Give the correct result for this request.

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Preserve the first match and finite search

> Explain why a compatible last row can still break the contract.

Your evidence record

- For the activity catalogue, 15 minutes with Quiet required No allows several activities. Explain why returning the last compatible row breaks the agreed rule even though it fits.

  Describe a stopping rule that preserves the first match, and one way to check that a loop cannot repeat forever.

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Design a complete room decision process

> Keep one first attempt before hints or feedback.

Read and explain independently

A new paper catalogue lists C1 first, capacity 4, quiet Yes; C2 second, capacity 8, quiet No. Valid requests give a whole group size from 1 to 10 and Quiet required Yes or No.

A room fits if capacity is at least group size and the quiet condition is met; choose the first fit. Write a complete **process** with **validation**, search, **state** change and stopping rules.

Trace 6 people No and explain the outcome for a blank quiet field. No booking takes place.

> **My independent explanation, trace and limits**

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Design a complete room decision process

MAIN LESSON · SECTION 9

Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

Your evidence record

Complete the sorter v1 **decision flow** and annotate every **branch** with the **evidence** it uses.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Track the current **candidate** for Designing a complete decision **process**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for Designing a complete decision **process**. Proposed **explanation**: “One failed row proves no match”. Decide whether this **explanation** is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

Rule: if paper then P, otherwise M. **Input**s are paper, metal, glass and unknown. Repair the rule for paper and metal trays without inventing a glass tray. Trace all four **input**s.

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

A rule sends paper to P and everything else to M. **Input**s include paper, metal and glass.

Trace glass and repair the rule.

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

Write a decision **process**: a clear pencil goes to the pencil tray, a clear book to the shelf, and an unknown object to review. Trace a lunch box.

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 31 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Reference · Original target response space**

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

Rule: if paper then P, otherwise M. **Input**s are paper, metal, glass and unknown. Repair the rule for paper and metal trays without inventing a glass tray. Trace all four **input**s.

> **My explanation after practice**

Record support used: none, hint or teacher **explanation**.

▼ Reference · Independent-case response copy

REFERENCE · SECTION 15

Reference · Independent-case response copy

> This duplicates W4. Reuse the same first-attempt record.

Use this when helpful

A new paper catalogue lists C1 first, capacity 4, quiet Yes; C2 second, capacity 8, quiet No.
Valid requests give a whole group size from 1 to 10 and Quiet required Yes or No.

A room fits if capacity is at least group size and the quiet condition is met; choose the first fit.
Write a complete **process** with **validation**, search, **state** change and stopping rules.

Trace 6 people No and explain the outcome for a blank quiet field. No booking takes place.

> **My reasoning and evidence**

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

> Keep the explanation and the support you used.

Use this when helpful

A fictional worksheet routes craft kits by required glue and number of scissors.

K1: glue No, scissors 2. K2: glue Yes, scissors 2. K3: glue Yes, scissors 4.

Request: glue required Yes, scissors needed 2. Return first kit meeting both requirements.



Follow the labelled picture with the matching explanation. Open the vector diagram

Where does the first original rejection occur, and why is K3 not originally selected?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Draw the sorter's decision **branch**es, loop-back arrows, final **output**s and invalid-**input** path so every valid case can terminate.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Helpful Classroom Sorter

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

Draw the sorter's decision **branch**es, loop-back arrows, final **output**s and invalid-**input** path so every valid case can terminate.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional worksheet routes craft kits by required glue and number of scissors.
- Which labelled step matches this detail? Return K2 without replacing it with later K3.
- 1. Stop
- 2. Evaluate K1
- 3. Evaluate K2

- 4. Validate fields

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported **explanation**: Stop Return K2 without replacing it with later K3. A complete decision **process** specifies what to do after each result. One passing requirement is insufficient.

After a kit passes both, stopping preserves the first-match contract; automatically choosing the largest later kit would implement a different rule. A full scan with no accepted kit returns no match.

Changed-case question

- Changed-case practice: Increase scissors needed to 3. Which conclusion is supported by the supplied facts?

- 1. K3 is selected.

- 2. We can decide the result without examining the changed **input** or the procedure.

- 3. This one example guarantees correct results for every future **input**.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported **explanation**: K3 is selected. K1 lacks glue; K2 has only 2 scissors; K3 has glue and 4.

Repair and save your work

Review Choose **Input**s and **Output**s (ai-1/30). Goal: Turn the project outcome into precise **input** and **output** requirements. Short practice: Trace requests 10 and blank. Why should they not receive the same **explanation**?.

Return to this week and explain what you changed.

Review Humans Set Boundaries (ai-1/28). Goal: Distinguish a system's capability, the correctness of its **output** and permission to take an action. Short practice: Which original proposal is authorised?

Why do the other two need separate approval?. Return to this week and explain what you changed.

Review Generate, Retrieve, or Calculate? (ai-1/25). Goal: Distinguish retrieving a record, calculating from **input**s and generating wording. Short practice: Label each arrow with its operation.

Can the system add snacks because invitations often mention them?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Design the Decision Process: the case

AI ACADEMY / AI-1/31 · 01 · LOOK AT THE FACTS

Design the Decision Process: the case

The situation

A fictional worksheet routes craft kits by required glue and number of scissors.

Supplied evidence

K1: glue No, scissors 2. K2: glue Yes, scissors 2. K3: glue Yes, scissors 4. Request: glue required Yes, scissors needed 2. Return first kit meeting both requirements.

By the end, I can design a complete decision flow with validation, search and stopping.

Follow the labelled picture with the matching explanation. Open the vector diagram

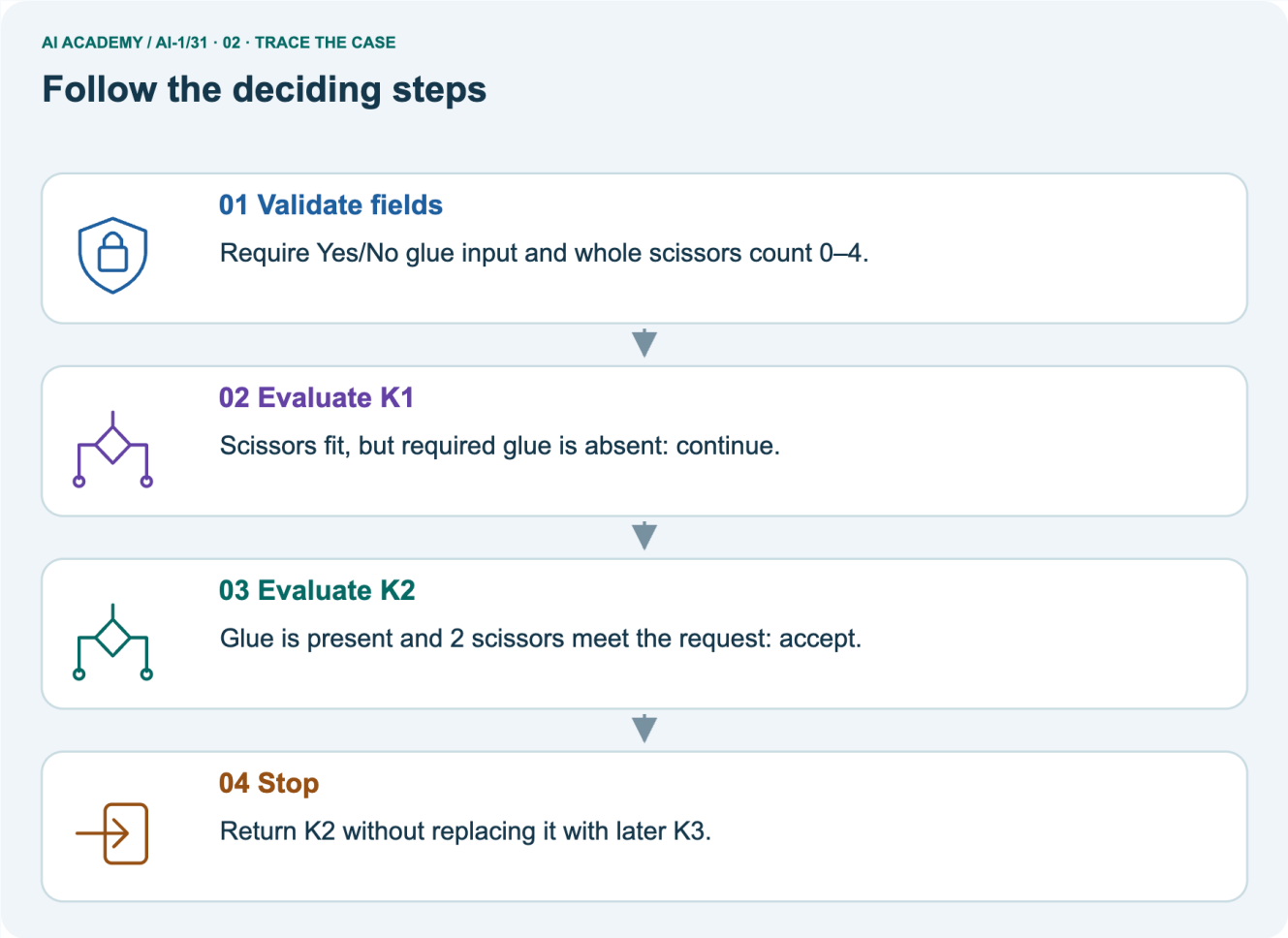
A fictional worksheet routes craft kits by required glue and number of scissors. Supplied facts: K1: glue No, scissors 2. K2: glue Yes, scissors 2. K3: glue Yes, scissors 4. Request: glue required Yes, scissors needed 2.

Return first kit meeting both requirements.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



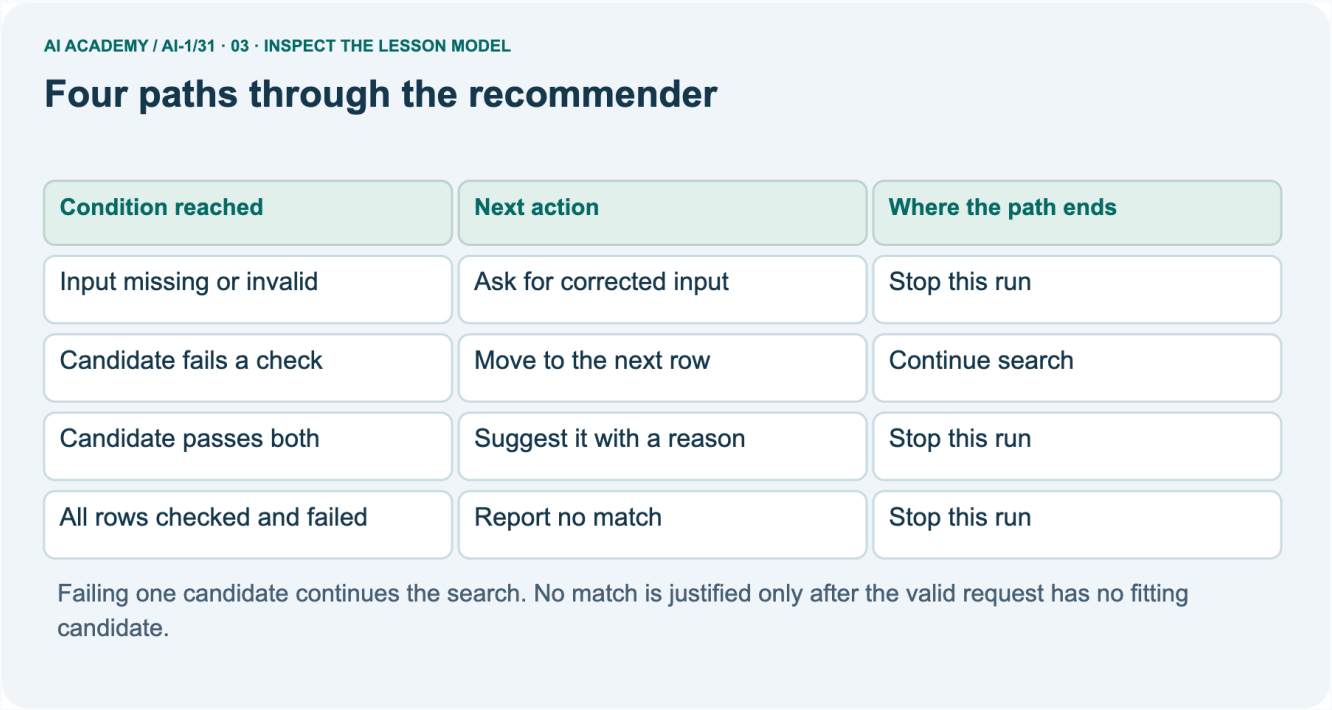
Follow the labelled picture with the matching explanation. Open the vector diagram

Validate fields: Require Yes/No glue **input** and whole scissors count 0–4. → Evaluate K1: Scissors fit, but required glue is absent: continue. → Evaluate K2: Glue is present and 2 scissors meet the request: accept. → Stop: Return K2 without replacing it with later K3.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Four paths through the recommender



AI ACADEMY / AI-1/31 · 03 · INSPECT THE LESSON MODEL

Four paths through the recommender

Condition reached	Next action	Where the path ends
Input missing or invalid	Ask for corrected input	Stop this run
Candidate fails a check	Move to the next row	Continue search
Candidate passes both	Suggest it with a reason	Stop this run
All rows checked and failed	Report no match	Stop this run

Failing one candidate continues the search. No match is justified only after the valid request has no fitting candidate.

Follow the labelled picture with the matching explanation. Open the vector diagram

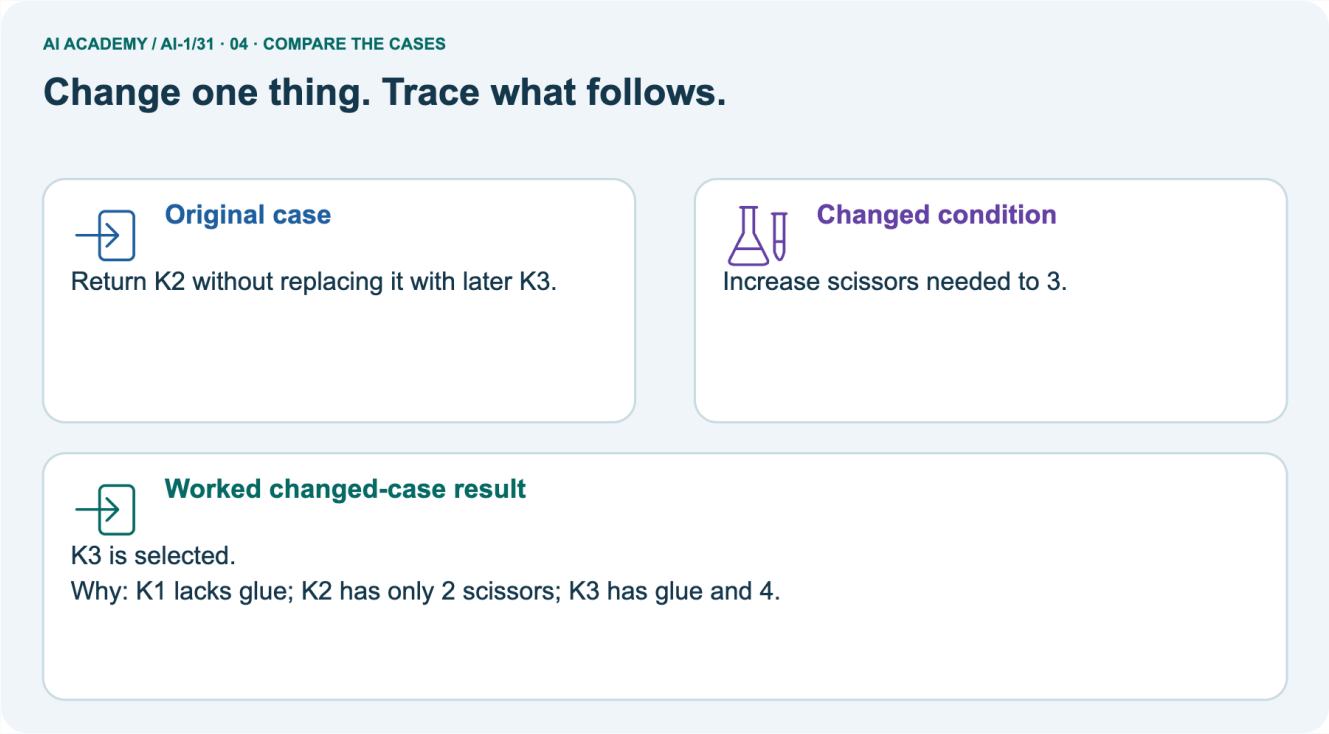
Four paths through the recommender. **Input** missing or invalid | Ask for corrected **input** | Stop this run. **Candidate** fails a check | Move to the next row | Continue search.

Candidate passes both | Suggest it with a reason | Stop this run. All rows checked and failed | Report no match | Stop this run

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



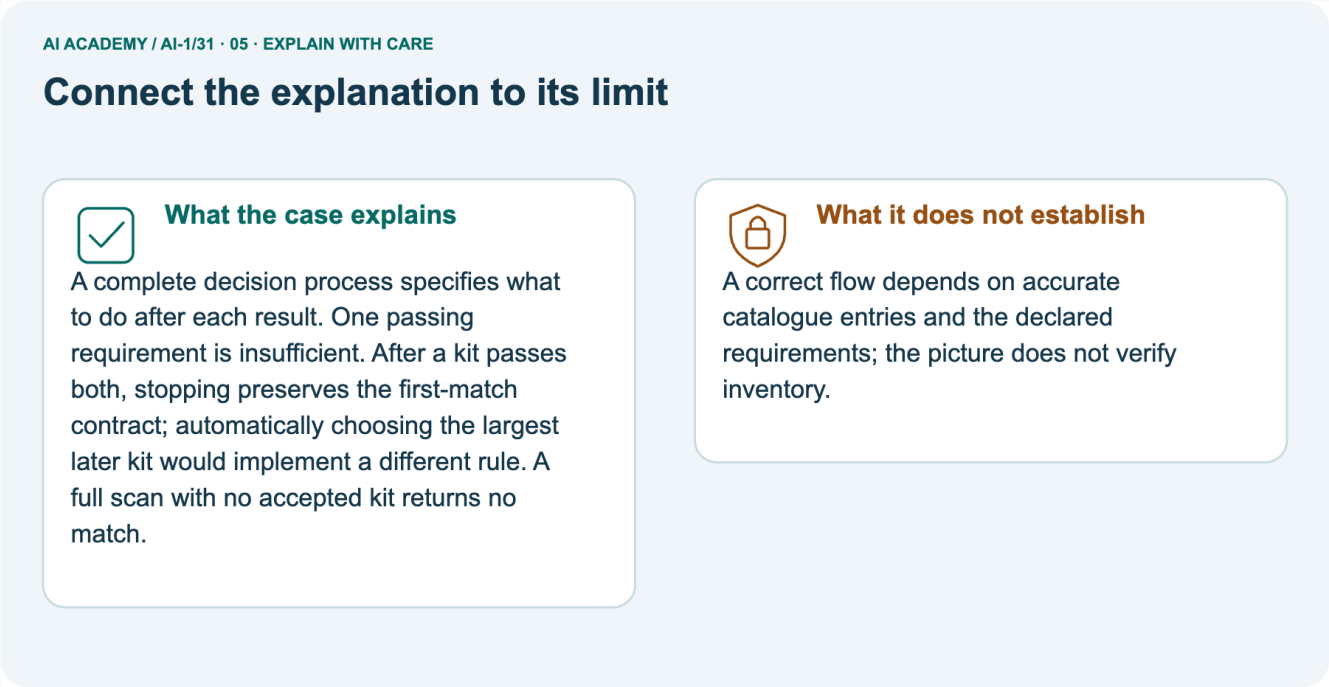
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Return K2 without replacing it with later K3. Changed condition: Increase scissors needed to 3. Expected result: K3 is selected. Reason: K1 lacks glue; K2 has only 2 scissors; K3 has glue and 4.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A complete decision **process** specifies what to do after each result. One passing requirement is insufficient.

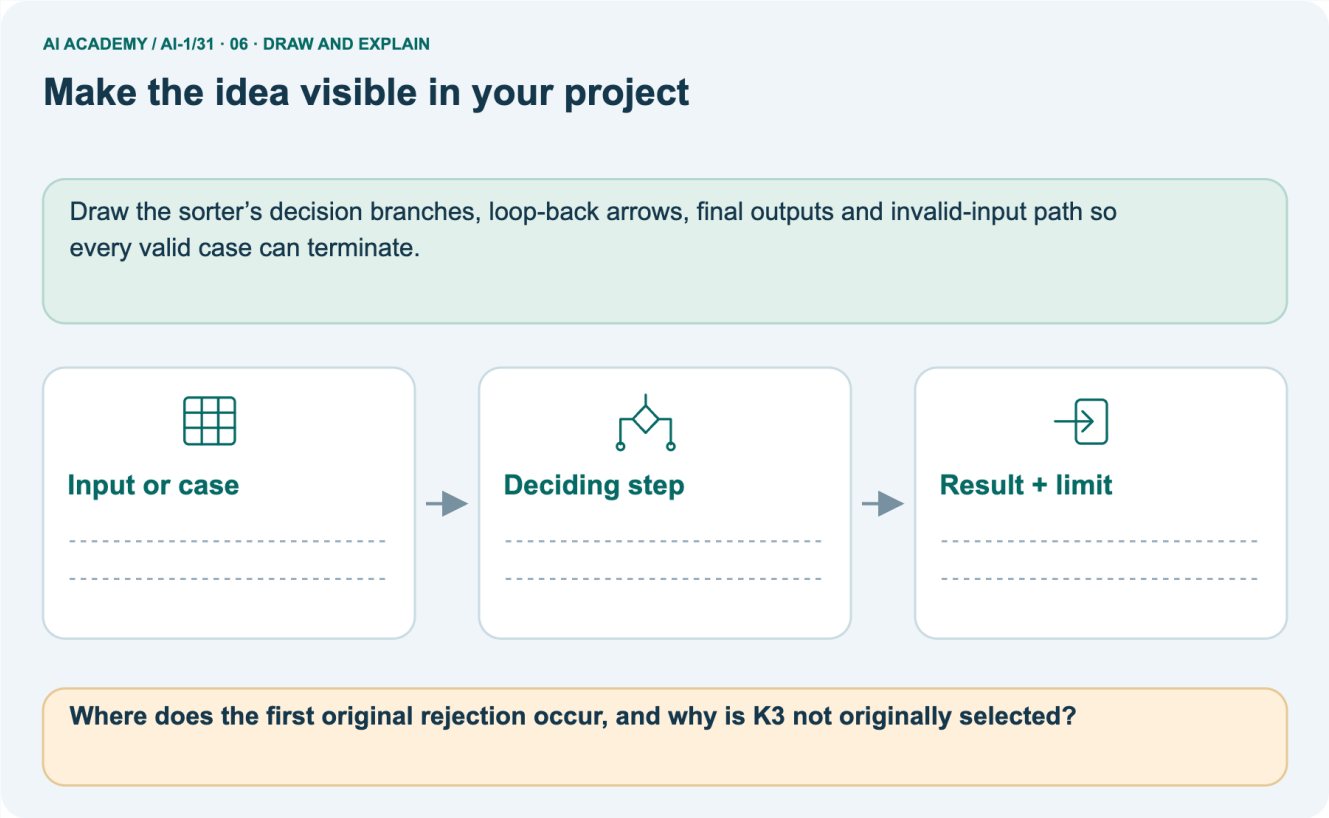
After a kit passes both, stopping preserves the first-match contract; automatically choosing the largest later kit would implement a different rule. A full scan with no accepted kit returns no match.

Limit: A correct flow depends on accurate catalogue entries and the declared requirements; the picture does not verify inventory.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Draw the sorter's decision **branch**es, loop-back arrows, final **output**s and invalid-**input** path so every valid case can terminate. Draw your **input**, deciding step, result and limit.

Supported question: Where does the first original rejection occur, and why is K3 not originally selected?

Draw your project trace and justify the deciding step using **evidence**.

> **My picture explanation, deciding evidence and one limit**

▼ Earlier wording, alternatives and picture transcripts

01. Design the Decision Process: the case

02. Follow the deciding steps

03. Four paths through the recommender

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.